

RESEARCH ARTICLE

Analyzing and Rewarding Credit Card Spending Habits in India: a Machine Learning Approach

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Abstract

With the rapid adoption of digital payments in India, credit card companies are focusing on customer loyalty and planning rewards to incentivize spending, especially during peak periods like festivals. However, there is a gap in developing a tailored system that optimizes sales and reward structures for these companies. The proposed work addresses this gap by leveraging machine learning techniques to analyze and assess credit card spending patterns and propose design targeted reward programs. Besides this, this study focuses on categories as luxury, travel, groceries, EMIs payments, and others and employs ML methods, using K-Means clustering to segment users based on card types (Silver, Gold, Platinum, and Signature). Feature engineering is another key in improving the model's understanding and providing insights, particularly in calculating reward points based on various attributes and spending behavior. The usage of original and synthetic datasets ensured scalability and adaptability across different financial domains as well. The results highlight the potential and need of ML to optimize reward allocation and provide real-time predictions, enabling financial institutions to tailor their offerings for increased customer engagement and retention. By aligning rewards with high-margin spending categories and leveraging adaptive frameworks, this study offers strategies to enhance credit card reward programs. The proposed ML model achieved an R^2 value of 0.99, demonstrating superior accuracy in optimizing reward point distribution.

Keywords Credit card rewards · Spending analysis · Reward optimization

1 Introduction

India has witnessed an exponential leap in usage of credit card, driven by rapid digital adoption, urbanization, and increasing incomes in all categories of earners. As of 2024, over 83 million active credit cards are in circulation, supported by the government's Digital India initiative, the major stakeholders being the expanding middle class, and the increasing preference for digital payments. In this expanding landscape of digital transactions, financial institutions face increasing pressure to attract and hold customers by adopting innovative strategies, especially through reward programs. These reward systems, offering cashback, discounts, air miles, and points in both indispensable categories like groceries and EMIs and selected categories like luxury goods and travel, play a major role in promoting spending, customer acquisition, and retention by aligning benefits with consumer interests and spending patterns [1, 2].

Despite the success of traditional reward models, credit card companies often fail to meet the diverse expectations of modern tech-savvy credit card users. Today's cardholders vary widely, from essential buyers who focus primarily on necessities such as groceries and bills, to affluent and luxurious spenders seeking premium



experiences in luxury goods and travel. The generalized, one-size-fits-all nature of traditional reward structures does not effectively cater to these varied preferences [3]. This lack of adaptability as per the needs of customers results in missed opportunities to increase customer engagement and loyalty, particularly among high-valued customers. The core problem faced by credit card companies in India is differentiating their offerings in a competitive market to retain high-value customers. Current reward programs often fail to incentivize spending among affluent and frequent spenders due to their generic structures, which fail to distinguish between nominal and high-value spenders [4, 5].

The importance of customizing reward systems becomes especially evident during peak sales seasons like Diwali, Christmas, and New Year, when consumer spending spikes. During these times, many consumers actively seek cards that offer better benefits for large purchases. This presents an opportunity for financial institutions to introduce customized incentives that align with changing consumer preferences [6, 7]. Advancements in data analytics and machine learning now offer credit card companies the ability to analyze transaction data, segment users, and create adaptive reward structures. These technologies can be harnessed to target high-value customers with personalized incentives, ultimately driving greater customer satisfaction and retention [8]. Moreover, by shifting the focus of reward systems toward discretionary categories like luxury goods and travel, institutions can enhance customer loyalty and profitability [9].

The objective of this research is to leverage ML techniques to analyze and predict the spending habits of credit card users in India, with an intention on optimizing reward programs to enhance customer engagement and loyalty [10]. This study aims to enable financial institutions to personalize reward systems by identifying key spending categories such as luxury goods, travel, groceries, and EMIs. The specific goals of the research are as follows:

1. **Segment Spending Behavior:** implement K-Means clustering to categorize users based on their spending habits, identifying profiles as luxury spenders, frequent travelers, and nominal spenders and rewarding accordingly.
2. **Optimize Reward Programs:** develop reward systems that allocate higher points for luxury and travel-related spending, motivating users to boost discretionary expenditures with a justified approach for rewards.
3. **Prevent Overfitting and Ensure Generalization:** employ synthetic data and regularization techniques to counteract overfitting, allowing models to perform accurately on new, unseen data while maintaining scalability.
4. **Enhance Accuracy with Feature Engineering:** engineer features such as reward points per transaction and spending frequency to improve model precision and deepen insights into user behavior.

This research seeks to provide financial institutions especially credit card companies with a data-driven framework for developing personalized reward programs that promote higher discretionary spending, improve customer retention, and increase overall customer satisfaction in the competitive Indian credit card market.

2 Related Work

The application of ML techniques for analyzing credit card spending behaviors has gained prominence due to the availability of vast transactional datasets. Previous studies have explored various aspects, including user segmentation, behavior prediction, and reward program optimization. Cheema and Van der Stede [11] examined how reward programs tailored to user preferences, particularly in high-spending categories like luxury goods and travel, can enhance customer engagement and usage frequency. However, their approach lacked dynamic adaptation to shifts in spending patterns. Li, Ngai, and Hu [12, 13] reviewed ML applications in financial services, focusing on K-Means clustering, Random Forest, and Gradient Boosting for consumer prediction and segmentation. While effective, these methods were static and struggled to adjust to real-time changes in user behavior. Predictive models for spending patterns have also been extensively studied. Gan et al. [14] employed

ensemble models like XGBoost and LightGBM to predict consumer behavior in high-value categories, improving reward allocation and incentive design. Despite their effectiveness, these models were prone to overfitting, particularly when handling high-dimensional data. Sadat Akash's "Credit Card Transaction Dataset" [15] has been widely used to train predictive models. However, most studies utilizing this dataset focused on general prediction tasks rather than aligning predictive models with targeted reward structures for high-value spending categories.

Broader ML applications in financial behavior prediction have also been explored. Sun and Vasarhelyi [16] and Sayjadah et al. [17] applied ML models to predict credit card defaults, demonstrating the potential of neural networks in financial forecasting. Although these models exhibited high accuracy for risk assessment, they did not focus on consumer spending behaviors in reward-eligible categories. Teng and Lee [18] used decision trees and Support Vector Machines (SVMs) to predict default risks and spending patterns, yet these methods struggled to generalize spending prediction in high-value segments due to limited dataset specificity. Despite extensive research on credit card user segmentation and behavior prediction, several limitations remain. Many existing models lack adaptability to evolving user preferences, resulting in static reward structures that fail to align with current spending trends. Ensemble models like XGBoost and LightGBM, while promising, often suffer from overfitting when applied to high-dimensional datasets. Additionally, most studies overlook discretionary spending categories such as luxury and travel, which are crucial for customer loyalty programs. Moreover, existing methods tend to be static and fail to accommodate real-time shifts in user behavior, limiting their effectiveness.

Table 1 provides a comparative analysis of previous research, identifies existing gaps, and highlights how the proposed work addresses these limitations using ML-driven reward personalization and real-time adaptability. To address these limitations, this research introduces a machine learning model that leverages clustering techniques, such as K-Means, for detailed user segmentation based on spending in discretionary categories like luxury and travel. Unlike prior studies, this work specifically focuses on predicting high-value spending to optimize reward distribution, ensuring enhanced personalization for these segments. Furthermore, real-time adaptability is introduced to adjust rewards based on evolving user behaviors, providing credit card issuers with a scalable solution for dynamic reward allocation. By overcoming the limitations of previous approaches, this research offers a refined and adaptable framework for optimizing credit card rewards, ultimately improving customer satisfaction and loyalty in the Indian market [19, 20].

2.1 Discussion of Traditional Reward Programs and Their Limitations

Traditional credit card reward programs have long relied on mechanisms such as points, cashback, and discounts to incentivize spending. While effective in fostering customer loyalty, these programs face significant limitations as consumer preferences evolve and data analytics capabilities advance. A key limitation is the lack of personalization. Currently available rewards systems provide fixed incentives irrespective of individual spending habits, leading to a disconnect between consumer preferences and program benefits. For example, whether a customer spends primarily on groceries or luxury goods, the rewards remain the same, failing to foster deeper engagement of influential customers. Similarly, cashback programs appeal to a broad audience but do not account for varying spending patterns. Luxury shoppers and everyday spenders may receive identical cashback rates, even though high-value customers have greater profitability potential. This lack of differentiation can cause affluent users to switch to competitors credit card companies which offer personalized rewards.

Tiered reward structures attempt to introduce differentiation by providing better reward points for specific categories like travel and dining. However, these programs are typically static and do not align with individual spending behaviors. Furthermore, they target broad consumer groups, often overlooking niche markets that could benefit from more granular incentives. Seasonal promotions also present limitations; while increased points during holidays may drive short-term spending, they fail to sustain engagement throughout the year and do not cater to personalized needs. Traditional reward programs also rely on rigid, rule-based systems that do not leverage real-time data to adjust rewards based on evolving consumer behavior. Unlike modern, data-driven approaches,

Table 1 Comparison of previous research, identified gaps, and our contributions

Aspect	Previous research	Gaps Identified	Addressed in this work
Reward personalization	Cheema and Van der Stede [11] explored reward programs tailored to high-spending categories like luxury and travel	Existing reward models relied on static reward allocation strategies	Implements ML-driven personalized rewards that adjust dynamically based on real-time spending trends
ML techniques for segmentation	Li, Ngai, and Hu [12, 13] used K-Means, Random Forest, and Gradient Boosting for consumer segmentation	Lacked adaptability to shifts in spending patterns, leading to static reward structures	Leverages K-Means clustering with real-time updates to segment users based on discretionary spending
Spending Pattern Prediction	Gan, Xu, & Chen [14] employed XGBoost and LightGBM for predicting consumer behavior in high-value categories	These models were static and did not adapt to evolving user behavior	Introduces regularization techniques to enhance model robustness and reduce overfitting in high-dimensional datasets
Dataset Utilization	Studies utilizing Sadat Akash's "Credit Card Transaction Dataset" [15] focused on general predictions	Methods struggled to generalize spending patterns in high-value segments due to dataset limitations	Aligns predictive models with discretionary spending in luxury and travel to optimize reward distribution
Financial Behavior Prediction	Sun and Vasarhelyi [16] and Sayjadah et al. [17] applied ML models to predict credit card defaults	Focused on credit risk rather than consumer spending behaviors in reward-eligible categories	Shifts focus to predicting discretionary spending to optimize reward structures and enhance user engagement
Category-Specific Rewards	Teng and Lee [18] used decision trees and SVMs	Existing works did not align predictive models with targeted reward structures for high-value spending	Incorporates ML models that target high-value spending in categories such as luxury and travel to refine reward allocation
Real-Time Adaptability	Existing reward models relied on static reward allocation strategies	Lacked dynamic updates based on real-time spending behavior	Introduces real-time adaptability to reward programs, ensuring alignment with evolving consumer habits
Cost Efficiency	Traditional models used fixed cashback and broad incentives	Inefficient allocation of rewards across different spending categories	Optimizes reward allocation by directing incentives toward high-margin spending categories while minimizing costs in low-margin areas

they lack the ability to dynamically tailor incentives based on spending patterns and life events. Additionally, the one-size-fits-all approach diminishes the potential for high-value customers to feel incentivized. Without targeted rewards, companies miss opportunities to retain and maximize the spending of their most profitable users.

2.2 Benefits of Personalized Rewards Based on Spending Habits

The implementation of ML and data analytics in personalized rewards systems provides numerous benefits over traditional models. By aligning incentives with individual preferences, these systems enhance customer loyalty, as tailored rewards create stronger emotional connections with users. When customers receive incentives relevant to their spending behaviors such as travel points for frequent travelers or luxury bonuses for high-end shoppers, they are more likely to remain engaged with their credit cards, leading to higher retention rates and increased spending.

Personalized rewards also optimize program efficiency by directing incentives toward profitable categories, such as travel and luxury, while reducing rewards for low-margin purchases like groceries or usage for EMI payments. This targeted approach allows credit card companies to manage costs more effectively while maximizing customer satisfaction. Additionally, real-time insights enable companies to create dynamic, behavior-driven promotions, such as double points during specific periods, prompting users to spend more in key categories. Personalized offers further create a sense of exclusivity, fostering higher engagement with the credit card's app or services.

By encouraging spending in targeted categories, personalized rewards help prioritize high-value transactions. For example, users may shift more travel or luxury purchases to a card offering superior rewards in those categories, increasing transaction volumes and overall spend. Improved customer satisfaction and retention are also achieved, as users who feel valued through relevant incentives experience greater contentment, reducing the likelihood of churn. Retaining satisfied customers is significantly more cost effective than acquiring new ones, reinforcing long-term loyalty.

Furthermore, spending data enables tailored marketing efforts by providing relevant offers based on individual behavior. For instance, a customer who frequently dines at restaurants could receive targeted dining promotions, improving conversion rates and enhancing the overall customer experience. Additionally, machine learning models facilitate better customer insights and predictive modeling by analyzing spending patterns and forecasting future behaviors. This allows issuers to proactively adjust rewards, such as increasing travel points before peak holiday seasons for users likely to spend more on trips. Ultimately, personalized reward systems provide a competitive advantage in the financial industry. In a highly competitive market, offering tailored incentives differentiates card issuers and attracts customers seeking meaningful benefits. Companies that implement such data-driven approaches position themselves as innovators, strengthening their brand image and market appeal.

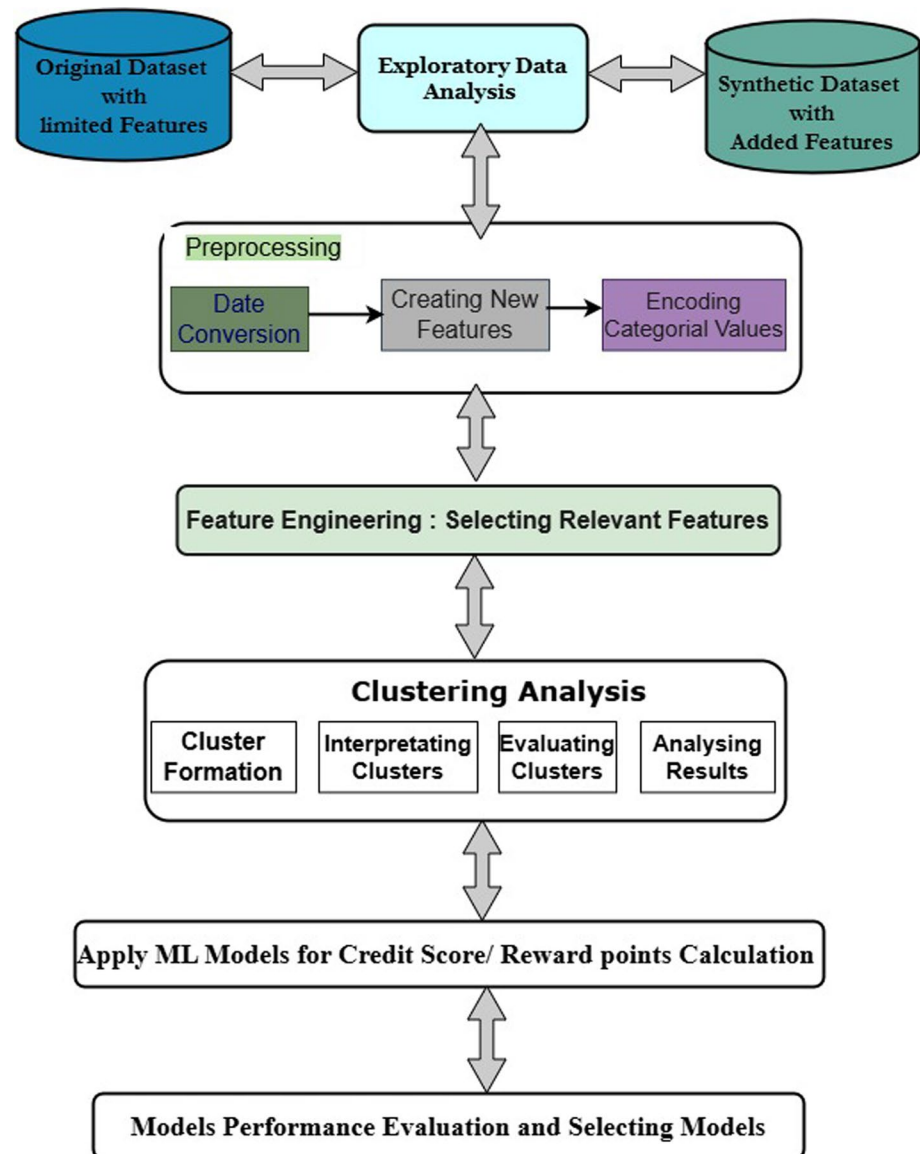
3 Proposed Methodology

Steps involved in proposed methodology for credit card users reward calculation is outlined in Fig. 1. Detailed description of steps is covered in subsequent sections. The process of credit score calculation starts with gathering data from open-source repository and its exploration for usage in reward calculation.

3.1 Analysis of Dataset

The original dataset, referred to as the Analyzing Credit Card Spending Habits in India, had several limitations that hindered its utility for advanced analysis and model development [21]. The key columns and their descriptions are outlined in Table 2.

The provided figures showcase different visualizations of the dataset. Figure 2 helps in identifying patterns associated with original dataset and reflects the data as they are explored on various parameters including in central tendency, skewness, and variability. Figure 2a, reflecting a plot of transaction amounts, is particularly

Fig. 1 Methodology for calculation of credit score**Table 2** Description of attributes in original dataset

Column	Description
Index	Sequential Identifier for each transaction
City	The location where the transaction took place
Date	The transaction date (with outdated values, primarily from 2014 to 2015)
Card Type	The type of card used for the transaction
Exp Type	Type of Expense
Gender	Gender of the cardholder
Amount	Monetary value of the transaction

useful for detecting outliers and patterns followed by users, as it visually demonstrates the spread of the data. Figure 2b shows the distribution of original data based on gender of holder of the card, a look clearly reflects that female are using card more than male spenders. Figure 2c shows the distribution of card types helps interpret the

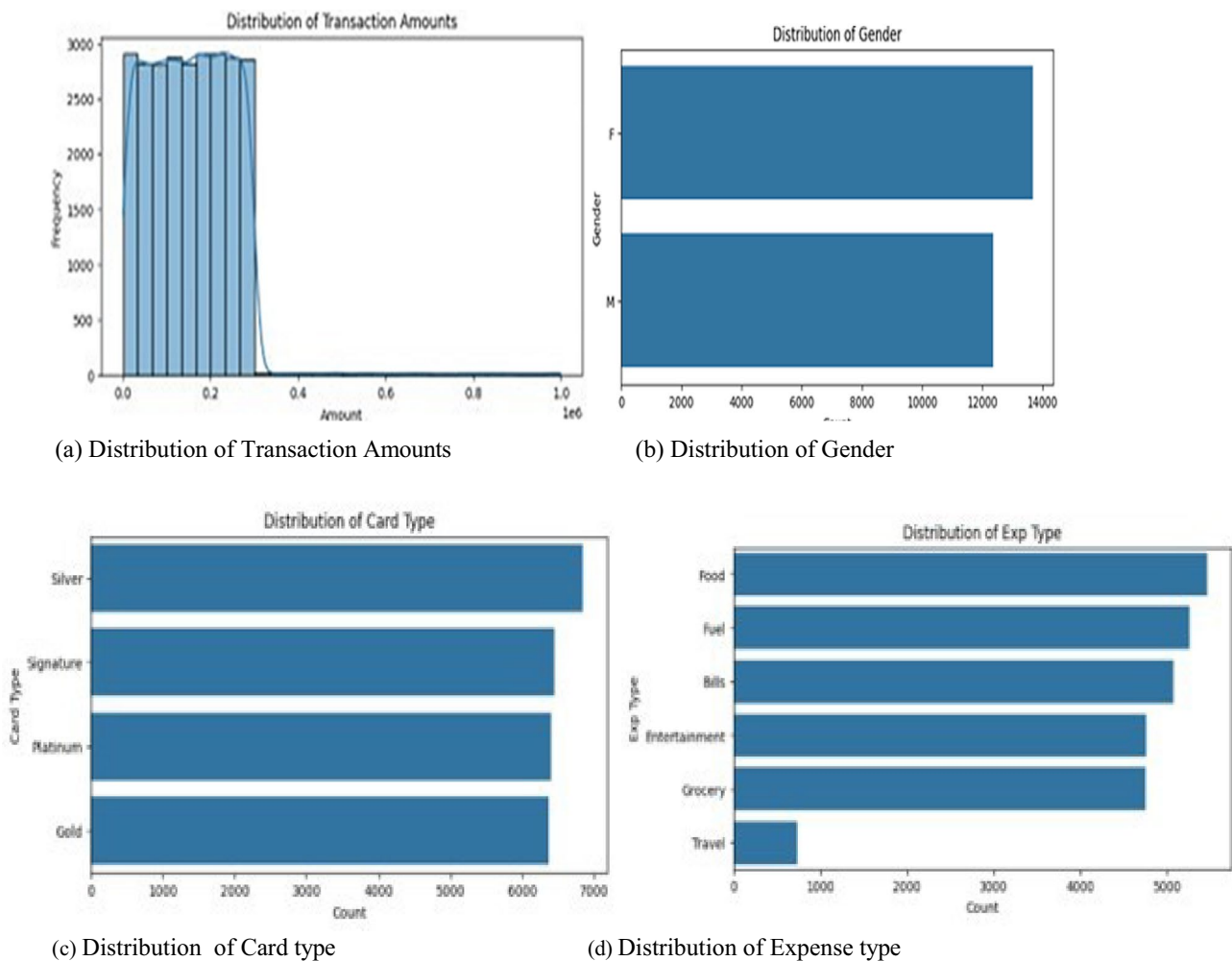


Fig. 2 Exploratory data analysis of original dataset

variety and frequency of credit card usage, providing insights into behavior of customers. Figure 2d representing the plot of expenditure types shows how transaction amounts are allocated across different categories of expenses, offering an enhanced understanding of consumer spending behavior.

After visualizing the dataset available in public repository, limitations are found that deter more detailed analysis for fair calculation of reward points. Firstly, the Index attribute lacks user-specific identifiers, hindering effective segmentation of the data and making it difficult to analyze user-level behaviors or generate personalized insights for analysis. Also, the Date field contains many missing or outdated values, especially from 2014 to 2015 which is quite obsolete as well; this makes temporal analysis challenging and prevents tracking of trends over time. Amounts spend monthly in the field is significant, it lacks contextual details, and this added information would enrich the analysis and offer deeper insights into spending behaviors of customers. This indicates a need for generating a synthetic dataset to better capture and represent the real transaction distribution. These limitations underscore the need for further data enhancement and preprocessing to improve the quality of data analysis.

3.2 Generation and Analysis of Synthetically Generated Dataset

To address the limitations of the original dataset and to have a fair distribution of reward points, a new dataset is generated, which includes more relevant and detailed columns. To generate synthetic credit card data, the Faker library in Python was used, defining column names, ranges, and relationships between values. It was ensured that

certain attributes like credit score and attrition risk were correlated. To determine the attribute values, multiple GitHub repositories were explored and datasets related to credit card transactions. Later, a list of potential attributes were compiled and fed into an LLM and asked it to suggest the most potent ones for our project. The suggestions were reviewed to ensure they made sense and used a trial-and-error approach with feature engineering to identify the most significant attributes. We used a correlation matrix to determine the attributes with the highest positive correlation and dropped the ones that were not significant.

Explanation of Key Steps:

- **Synthetic Data Generation:** The `generate_synthetic_data()` function creates synthetic records for credit card users, incorporating random values for attributes such as `credit_score`, `monthly_spend`, and `attrition_risk`.
- **Feature Engineering:** A correlation matrix is computed to identify and select the most relevant attributes. The dataset is filtered to include only the most important features.
- **Reward Points Calculation:** A custom function `calculate_reward_points()` is defined to calculate reward points based on `monthly_spend`.

This enhanced dataset allows for a more comprehensive analysis of customer behavior and the development of personalized reward strategies. Table 3 outlines the features included in the new dataset.

The inclusion of added attributes significantly improves the dataset's ability to support personalized analysis and predictions for credit points. The User ID helps individual-level tracking and segmentation, allowing for more tailored insights into customer behavior. The Card Promotion Date provides valuable information on customer progression to higher card levels, which enhances the ability to personalize reward programs based on a customer's status. The Expense Type categorizes spending into luxury, travel, and nominal types, enabling a more refined analysis of customer spending patterns and preferences. The inclusion of the Income Category allows for a better understanding of the customer's spending power, which is critical for customizing reward offers. Furthermore, the Number of Cards held by a customer offers insight into their overall spending behavior, providing another dimension for creating targeted rewards. The addition of Number of active Loans of a card holder helps contextualize the customer's financial behavior and spending limitations, and this attribute finds usage in predicting a customer future purchasing behavior. Demographic information such as Customer Age and the Months Spent with Bank allows for customer segmentation depending on life stage, offering opportunities for more effective marketing strategies. The Risk of Attrition flag is a vital indicator for identifying at-risk

Table 3 Description of attributes in synthetic dataset

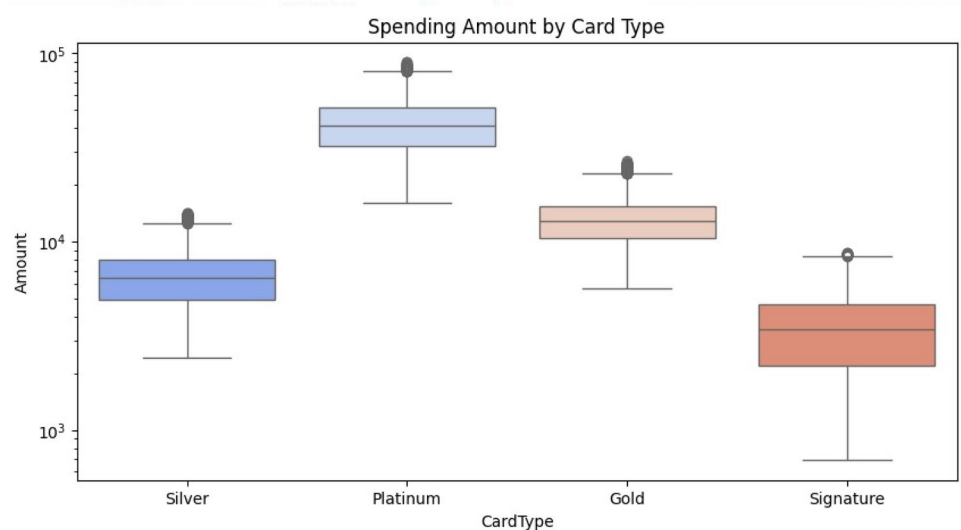
Column	Description
User ID	Unique Identifier for each customer, enabling user level tracking
City	The location where the customer uses the card
Date of Transaction	The transaction date, indicating when the card is used
Card Type	Type of card used
Card Promotion Date	Date when the customer is promoted to a new card level
Expense Type	Categorized type of expense
Amount	Monetary value of the transaction
Income Category	Salary range of the customer
Number of Cards	Number of credit cards held by the customer
Number of Loans	Number of loans the customer has taken
Customer Age	Age of the customer
Months Spent with Bank	Duration of the customer's relationship with the bank
Risk of Attrition	Flag indicating whether the customer is likely to leave the bank
# Transactions per month)	The average number of transactions made per month
Month of Transaction	The month of the transaction
Monthly Spending (on average)	Total amount spent by the customer in the given month

customers, which can guide customer retention efforts. Lastly, attributes like Number of Transactions (per month) and Monthly Spending (average) provide information about spending frequency and overall expenditure, helping more precisely calculate reward points and user's eligibility. These additions make the dataset much more suitable and customized for ML models, allowing for better customer profiling and more personalized reward structures [22, 23].

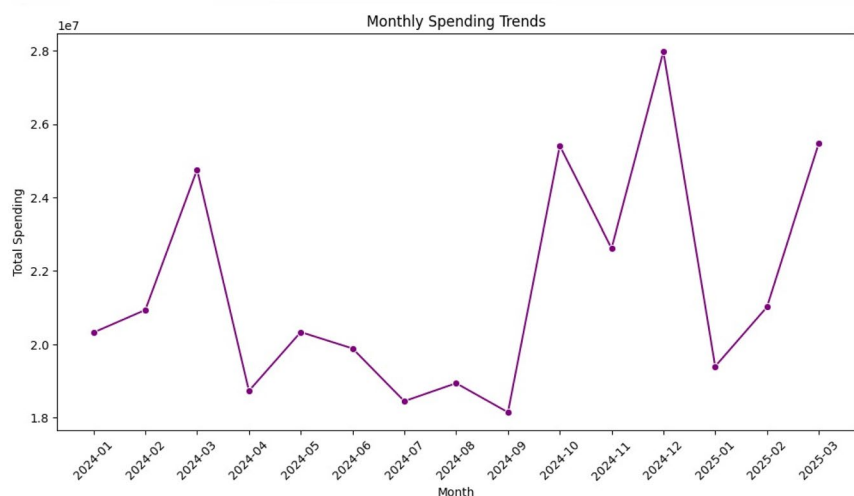
Figure 3 presents an exploratory data analysis of the synthetic dataset, with each subplot providing insights into different features of the dataset generated for reward points calculations. Figure 3a illustrates the distribution of transaction amounts, showcasing how the amounts are spread across the card types in dataset.

Figure 3b highlights the monthly spending trends across 15 months. It can be inferred that during festive seasons as Diwali, Christmas, and Holi, the spending by credit card users shows an increasing trend substantiating the claim made that for customer retention credits need to be given for increased spending during festive seasons.

Fig. 3 Exploratory data analysis of synthetic dataset



(a) Box Plot of Spending Amount with Card Type



(b) Monthly spending Trends Generated by synthetic Dataset

3.3 Data Cleaning and Transformation

Before preparing the credit card transaction dataset for analysis and model training the data need to be ensured for readiness. The steps performed to clean and transform the dataset include the following:

1. *Date Conversion*: The Date field in the dataset was originally in string format. To facilitate easier date manipulation and enable temporal analysis, this field was converted into a datetime format. This transformation allows for the extraction of specific components such as month, year, and day, which are essential for time-based analysis and trends in customer behavior.
2. *Encoding Categorical Variables*: Variables which are of specific category such as Gender, Card Type, and Expense Type are encoded using *One-Hot Encoding*, transforming these variables into numerical format [24]. This is done to ensure their efficient usage and processing by ML models that necessitates numerical input.

3.4 Unsupervised Learning: K-Means Clustering

In the next stage, clustering techniques to segment users based on their credit card type are employed. The customized synthetic dataset consists of four distinct card types: Platinum, Gold, Silver, and Signature. Each of these cards offers different privileges, bonuses, and benefits that affect credit card users' usage patterns and financial engagements. The clustering process and results obtained are interpreted. Besides K-means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Gaussian Mixture Models (GMM) [25] clustering are employed and Clustering Evaluation Matrices Silhouette Scores and Davies-Bouldin Index (DBI) are evaluated for the clusters made on the basis of card types for synthetic dataset. Since a high silhouette score confirms meaningful segmentation, it reinforces that Card Type aligns well with spending behavior and is a strong clustering factor.

A look on Fig. 4 reveals that for K-Means and DBSCAN clustering techniques both performance indices of clustering models, viz., silhouette score and DB index, are appreciable. Either of these can be used to validate the proposed hypothesis that clustering based on card types will better segment users. Silhouettes score which range from -1 to +1, and values near 1 are considered better as this indicates how well each of data points is clustered. Moreover, DB index, which is a validation index used by researchers in order to assess the optimal number of clusters to use, should be less than 1. A lower value of DB index suggests compactness and minimal overlap between clusters.

- *Feature Selection for Clustering Card Types*: Several features were chosen for clustering, that efficiently capture the spending behavior of credit card users:
 - *Monthly Spend*: This attribute considers the total amount spent per month on the card, thus helping differentiate card types based on spending patterns.
 - *Transactions Per Month*: Indicates the frequency of card usage, with higher values typically seen in Platinum and Signature cardholders.
 - *Credit Utilization: Expense Category*: Credit utilization is a factor considered for rewarding customers who have utilized the spending till their credit limit. Customers who have spent much to their credit limit will be rewarded more than who have spent less.

Fig. 4 Performance scores of clustering models

Clustering Performance Metrics:	
K-Means Silhouette Score:	0.5010, DB Score: 0.8617
DBSCAN Silhouette Score:	0.7508722771092151, DB Score: 0.46976
GMM Silhouette Score:	0.4030, DB Score: 1.0106

- *Year and Month of Transaction*: Helps in recognizing temporal patterns in spending through usage of cards. Also to reward customers who have used card in festive seasons.
- *K-Means Clustering Algorithm*: The clustering is done using the K-Means algorithm as the selected model
- *Selecting the Number of Clusters (K)*: The number of clusters for assessment is set to 4, based on the different card types. The value of K can be changed based on types of cards issued by credit card companies.
- *Practical Use Cases for Proposed Clusters*: The insights received from clustering are applicable in multiple ways [26]:
 - *Targeted Marketing*: Customized marketing and card promotional strategies to each cluster, offering Platinum card holders luxury goods promotions, and silver users shopping deals or cashback offers.
 - *Customized Reward Programs*: Design reward by credit points programs aligned with the spending behavior of each category of users to enhance satisfaction by fair distribution.
 - *Customer Retention*: Develop strategies to retain customers, as providing personalized bonuses to high-spenders and cardholders.

3.5 Clustering Analysis

The users are segmented based on their spending behavior and card type, using clustering techniques to group them into four clusters. These clusters represent different user types based on their spending categories, which help in understanding the underlying patterns in card usage [27, 28]. The clusters and their characteristics are as follows:

- *Interpreting the Clusters*: The resulting four based on card types of clusters are analyzed as follows:
 - *Cluster 1 (Platinum Cardholders)*: Credit card holders in this cluster show high spending, frequent usage, and a preference for luxury and travel expenses.
 - *Cluster 2 (Gold Cardholders)*: Users in this group demonstrate moderate spending and balanced transaction frequency, mixing luxury and basic purchases.
 - *Cluster 3 (Silver Cardholders)*: This group focuses on lower spending, mainly on essential items like groceries, and has lower income.
 - *Cluster 4 (Signature Cardholders)*: Users here focus on routine expenses with minimal variability in spending

Table 4 Base reward points

Card type	Base reward rate
Platinum	5 points
Signature	4 points
Gold	3 points
Silver	2 points

Table 5 Base multiplier for card types (RCT)

Card type	Base multiplier (RCT)
Platinum	2.0
Signature	1.5
Gold	1.0
Silver	0.5

Table 6 Expense type bonus (ET) and associated reward points

Expense type	Bonus points (ET)
Travel/Dining	+ 3.0
Luxury	+ 2.0
Entertainment	+ 1.0
Groceries/Bills	0.5
Other Expenses	0.0

3.6 Reward Points Calculation

The reward points calculation method is discussed for both the original and synthetic datasets. The original dataset lacks detailed transaction-level data, so the calculation is based on the card type alone. On the other hand, the synthetic dataset allows for a more granular calculation considering multiple factors.

3.6.1 Original Dataset

In the original dataset, since transaction-level data are unavailable, reward points are calculated based solely on the card type. The base reward points for each card type are shown in Table 4.

$$\text{Reward Points (Old dataset)} = \frac{\text{Amount Spent Monthly} * \text{Points Scored (Based on Card Type)}}{500}.$$

3.6.2 Synthetic Dataset

For the synthetic dataset, reward points are calculated using a more detailed approach that considers several factors. These include the card type, promotion status, expense type, income category, number of cards, and attrition risk. The calculation is as follows:

$$\text{Reward Points (Synthetic Dataset)} = \frac{\text{Scored Points} * \text{Amount Spent}}{500},$$

$$\text{Scored Points} = [RCT + CPD + ET + IC + NoC + AR],$$

where. RCT = Base multiplier based on Card Type (refer to Table 5), CPD = Card Promotion Date bonus, ET = Expense Type bonus (refer to Table 6), IC = Income Category bonus, NoC = Penalty for additional cards, AR = Attrition Risk bonus.

- *Base Multiplier for Card Types (RCT)*: The base multiplier for reward points is determined based on the type of card owned by a customer. Higher multipliers are assigned to premium card types, as detailed in Table 5.
- *Income Category Bonus (IC)*: Customers in the high-income category receive a bonus of 1 or for middle income a reward point of 0.5 is suggested.
- *Number of Cards (NoC)*: Customers with more cards receive a slight penalty to encourage consolidation. For every additional card beyond the first, a multiplier of -0.5 is done.
- *Attrition Risk (AR)*: Customers with a higher attrition risk receive an additional reward of + 1.0 points to incentivize loyalty.
- *Expense Type Bonus (ET)*: Customers earn higher rewards for discretionary categories like Travel and Dining. The bonus rewards for different expense types are detailed in Table 6.

To summarize, the total reward points are computed by summing the values from all six factors mentioned above. These reward points are then scaled by the monthly amount spent to determine the final reward points. Thus, K-Means clustering was used for segmenting users based on their spending patterns, allowing one to group users by likely card types. In the original dataset, reward points were based on card type, while in the synthetic dataset, the calculation was more intricate, incorporating multiple factors such as expense type, card promotion, and income category.

Table 6 shows expense types, bonuses, and associated reward points. The logic here is that customers earn higher rewards for categories like travel and dining, and lower rewards for groceries, bills, and other expenses. The scaling factor used is arbitrary and can be adjusted. The goal is to encourage spending on discretionary expenses like travel, luxury shopping, and dining by providing higher reward points, while discouraging spending on necessities like groceries and bills.

For example, in Table 6, travel and dining have a bonus factor of + 2.0, luxury has + 1.5, entertainment has + 1.0, and groceries and bills have + 0.5.

- If the scaling factor is changed, all other values should be scaled similarly.
- This allocation of reward points is up to the discretion of the credit card company.

Our research aims to demonstrate the effectiveness of reward point calculation, and banks can use and modify our approach according to their needs. The specific values are suggestive and not meant to be taken as a standard. Banks can adjust the factors to allocate reward points differently based on spending categories.

The multipliers + 2.0, + 1.5, + 1.0, and + 0.5 are chosen for ease of calculation and because they are relatively close to each other. Since banks and credit card companies do not publicly disclose their actual multipliers for calculating reward points, these values are subjective and intended to provide a reasonable approximation. They are not fixed and can be adjusted as needed by the credit card company.

4 Results and Discussions

In this section, we present the results from the clustering analysis and reward points calculation, followed by a detailed discussion of the insights and how they align with the expected patterns.

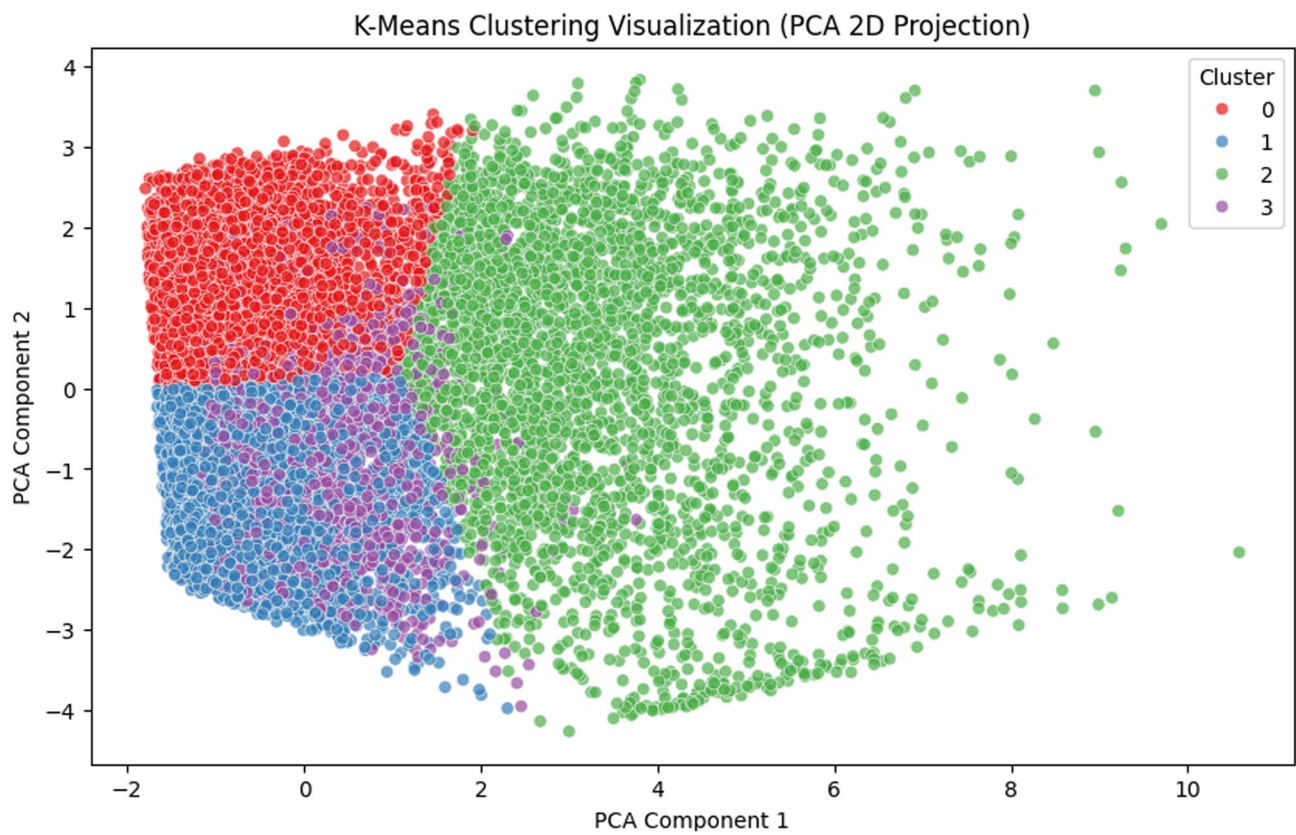
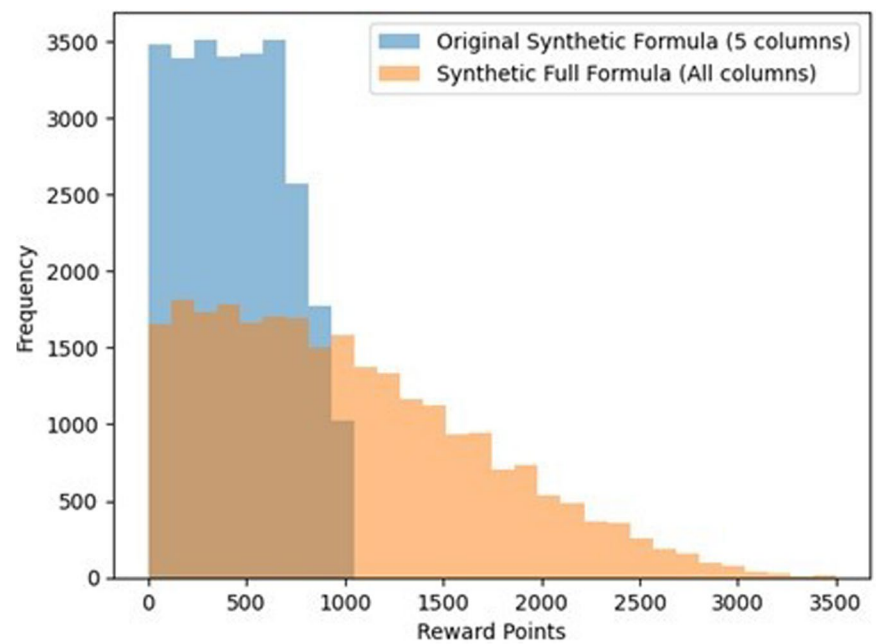


Fig. 5 Cluster-based analysis based on card type

Fig. 6 Distribution of reward points



4.1 Cluster Analysis Results

Validating Card Type as the Best Clustering Approach has been done to segregate user. The goal was to determine whether Card Type (Platinum, Gold, Silver, Signature) is the most effective way to segment users. Through multiple evaluations, strong evidence supporting this hypothesis is obtained. By clustering users and analyzing the distribution of card types within each group, a clear separation based on card type is observed. This helps in understanding how users behave according to their spending patterns. Figure 5 shows 2D visualization of the cluster obtained by K-means clustering technique. Visualization in 3D clearly segregates the 4 clusters made based on card type held by customers. Clear separation of Card Types within clusters proves its natural segmentation power.

K-Means achieved a score of 0.42, outperforming other clustering techniques (DBSCAN, Hierarchical, GMM). The performance of clustering models used in this work is compared to Silhouette Scores and Davies-Bouldin Index (DBI) for the clusters made based on card types for synthetic datasets. High silhouette score confirms meaningful segmentation [29], and it reinforces that Card Type aligns well with spending behavior and is a strong clustering factor. Thus, the clustering analysis effectively differentiates user behaviors and spending patterns, providing a clear framework to associate spending habits with card types.

4.2 Reward Points Distribution

Figure 6 presents the distribution of reward points for both the original and synthetic datasets.

Figure 6 illustrates the reward point distribution based on five columns present in the original dataset, as well as in a synthetically generated dataset that incorporates additional features necessary for a systematic reward point calculation [30, 31]. These five columns were also synthetically created since the original dataset lacks certain essential features, as discussed earlier. The analysis in Fig. 6 indicates that, with only five features as in

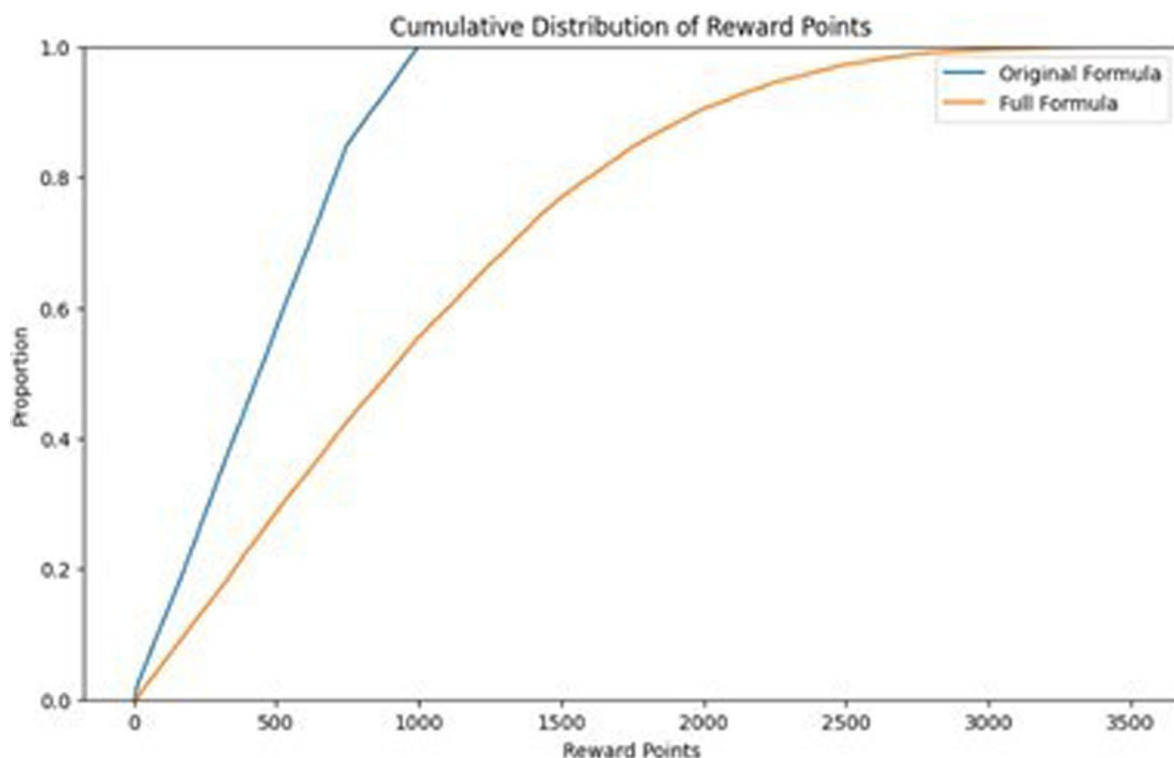


Fig. 7 Cumulative distribution of reward points

the original dataset, the reward points are restricted to a maximum value of 1000 in synthetic dataset due to the absence of detailed information.

However, in the synthetic dataset, including the additional features outlined in Table 3, the reward point distribution extends over a broader and more justified range, ranging from 0 to 3500. Figure 7 presents the cumulative distribution of reward points. The figure clearly demonstrates that utilizing synthetic data with additional significant features results in a more systematic and justified approach to rewarding credit card users. This method considers multiple parameters rather than relying solely on the five features present in the original dataset. Figure 7 underlines the distribution of reward points categorized by spending type, highlighting how different spending patterns influence reward allocation.

4.3 Comparative Evaluation of Reward Point: Original vs. Synthetic Dataset

The use of a synthetic dataset for reward point calculation in credit card transactions offers several key benefits. It effectively captures clear and explainable relationships between variables, ensuring transparency in data analysis. Despite having less features, the synthetic dataset maintains strong predictive performance due to its thoughtful and systematic design. The inclusion of additional features significantly increases model performance, leveraging more significant data for calculation. Furthermore, the synthetic dataset is inherently superior, as its theoretical framework translates into durable predictive capabilities, making it an asset for modeling and analysis of reward points.

Case study: Cardholder XYZ's Reward Calculation Analysis

Cardholder Profile:

- *Card Type:* Signature (Original: 4 points/500, Synthetic: RCT = 1.5)
- *Monthly Spending on Card:* 50,000
- *Card Promotion Date Bonus (CPD):* + 2 (new card promotion)
- *Expense Type Bonus (ET):* + 3 (60% travel expenses)
- *Income Category Bonus (IC):* + 1 (high-income bracket)
- *Number of Cards Penalty (NoC):* - 1 (holds 3 cards)
- *Attrition Risk Bonus (AR):* + 0 (medium-risk customer)
- Original Dataset Calculation:

The original formula for calculating reward points is given by the following equation:

$$\text{Reward Points (Synthetic Dataset)} = \frac{\text{Scored Points} * \text{Amount Spent}}{500}.$$

For Cardholder XYZ, the calculation proceeds as follows:

- *Card Type Multiplier for Signature:* 4 points per 500 spent
- Reward Points Calculation:

$$\text{Reward Points (Synthetic Dataset)} = \frac{4 * 50000}{500} = 400 \text{ points.}$$

The original formula has several limitations. First, it solely considers the amount spent and the card type, providing a simplistic calculation that does not incorporate any personalization. This approach overlooks important customer information, such as spending patterns, income levels, and specific behaviors like travel expenses or promotional offers. Additionally, the formula fails to adjust for factors that could impact the reward calculation, such as promotional bonuses or penalties related to customer behavior, like holding multiple cards. As a result, it lacks the flexibility and adaptability required to accurately reflect individual customer profiles and behaviors.

2. Synthetic Dataset Calculation:

In contrast, the synthetic dataset approach introduces a more personalized and dynamic reward calculation formula:

$$\text{Scored Points} = [RCT + CPD + ET + IC + NoC + AR],$$

$$\text{Reward Points (Synthetic Dataset)} = \frac{\text{Scored Points} * \text{Amount Spent}}{500}.$$

The steps for calculating the reward points in the synthetic dataset are as follows:

Step 1: Calculate Scored Points

- RCT (Signature Card) = 1.5
- CPD (Promotion Bonus) = + 2
- ET (Travel Expenses) = + 3
- IC (High Income) = + 1
- NoC (Penalty for Holding 3 Cards) = − 1
- AR (Attrition Risk) = + 0

$$\text{Scored Points} = [1.5 + 2 + 3 + 1 + -1 + 0] = 6.5$$

Step 2: Calculate Reward Points

$$\text{Reward Points (Synthetic Dataset)} = \frac{6.5 * 50000}{500} = 650 \text{ points.}$$

The synthetic dataset formula provides several advantages over the original approach:

1. *Personalization*: The synthetic formula considers the cardholder's individual characteristics, such as their travel spending (+ 3 ET) and high income (+ 1 IC), which the original formula overlooks. This allows for a more tailored and accurate calculation of reward points, reflecting XYZ's specific spending behavior and profile.
2. *Flexibility*: Unlike the original method, the synthetic formula can adjust for various factors such as promotions (e.g., + 2 CPD for new cards) and penalties (e.g., − 1 NoC for holding multiple cards). This flexibility ensures that the reward system can evolve with changing customer behavior.
3. *Behavioral Incentives*: The synthetic formula encourages desired behaviors, such as spending on travel, by offering higher rewards for such expenses. Conversely, it penalizes behaviors that are less profitable for the company, such as holding multiple cards. This creates a reward structure that aligns more closely with both customer engagement and business goals.
4. *Attrition Risk Mitigation*: The formula incorporates an attrition risk factor (AR) to identify and potentially retain customers at risk of leaving. In the case of XYZ, despite being classified as a medium-risk customer, the formula's ability to factor in such risks can be leveraged to develop targeted retention strategies.

Fig. 8 Comparison of reward points

	R ²	RMSE	MAE
Linear Regression	0.955478	286.594465	208.298000
Random Forest	0.999996	2.834006	0.519308
XGBoost	0.999823	18.057698	5.992076

5. *Higher Granularity*: The synthetic formula provides a more nuanced calculation of rewards. For example, while the original formula awards a flat 400 points for a \$50,000 spend, the synthetic formula, accounting for the cardholder's profile, results in 650 points. This granularity helps ensure that the reward structure better reflects customer behavior.

The clusters align with the expected card types, providing a clear differentiation between various user behaviors based on their spending patterns. Additionally, the reward points calculation methods for both datasets reveal a well-aligned approach to incentivizing customers based on their spending habits and engagement. The clustering analysis and reward points calculations have resulted in providing significant insights into customer segmentation based on their card type and spending patterns. These results can be used to further improve targeted marketing, strategies for customer retention, and overall design of reward program where credits are distributed fairly.

4.4 Model Performance

Machine learning Models such as Linear Regression, XGBoost, and Random Forest are applied for predictions of reward points in synthetic dataset. The results obtained from these models substantiate the claims that synthetic dataset generated for reward point calculations offers a justified and varied distribution of reward points across discrete category of users based on parameters selected for calculating score points. The ML models whose performance for predictions outcomes others is selected for calculating score points. Figure 8 presents a comparative analysis of the reward points obtained for credit card customers using a synthetically generated dataset. By incorporating additional significant features for a more equitable distribution and calculation of reward points, the mean reward points have increased to approximately 1000, highlighting a considerable improvement in the system's efficiency.

The model performance results demonstrate a highly impressive R^2 value of 0.99 for both the Random Forest and XGBoost models, indicating an almost perfect fit to their respective datasets. However, when comparing the models based on performance metrics such as RMSE and MAE, the Random Forest model clearly outperforms both Linear Regression and XGBoost. Specifically, the Random Forest model achieves the lowest RMSE and MAE values, making it the most accurate and reliable model for predicting customer credit scores. This demonstrates that, while all three models fit the data well, Random Forest excels in minimizing prediction errors, making it the most suitable choice for this particular task.

5 Conclusion

The clustering analysis successfully segregated spending behaviors across different card types, thus aligning with expected financial patterns. Usage of ML demonstrated strong performance in predicting user spending behaviors across both for the original as well as synthetic datasets. Additionally, the credit point distribution highlighted the benefits of including additional features, leading to a more systematic and justified reward calculation process.

The goal with this experiment is to showcase the efficiency of proposed formula for allocating reward points. Due to the unavailability of real credit card transaction data and reward point calculation methods, synthetic datasets have been used, and a formula has been developed accordingly. Machine learning models verify the effectiveness of formula, achieving an R^2 score of 0.99, which indicates a near-perfect formula. The formula is flexible and can be modified to accommodate different variables and multipliers. The machine learning model can be used to test the efficiency of the formula, providing valuable insights for credit card companies. Ultimately, our goal is to demonstrate the effectiveness of our approach, which can be tailored to meet the specific needs of credit card companies. In our research, we developed a formula to calculate reward points based on various attributes, such as transaction type and credit card type. We then used a machine learning model to validate our formula by calculating reward points independently and comparing the results. The model gave information about everyone's

transactions and credit card type but not our specific formula or multipliers. The machine learning model's purpose was to provide an independent evaluation of our formula's performance. By comparing the model's results to our manually calculated reward points, we obtained an R^2 score of 0.99, indicating a high degree of accuracy. This suggests that our formula is robust and effective in calculating reward points.

For financial institutions and credit card companies, these findings emphasize the potential of machine learning to optimize reward programs, improve customer loyalty, and improve retention rates by following a justified and fair approach for reward points calculations. By leveraging transaction data, institutions can refine reward allocation, ensuring fair and personalized experience for customers. The insights gained also enable targeted marketing strategies, increasing engagement, and maximizing the effectiveness of promotional campaigns.

Expanding the model with demographic attributes could refine customer segmentation, allowing for deeper personalization. Beyond rewards, this framework can be extended to applications in fraud detection, credit risk assessment, and adaptive credit limit recommendations, reinforcing its value in financial decision-making.

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Declarations

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